香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

Lecture 25: Oscillation, Wave

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April 22nd, 2025

Announcement

- 4 more lectures (including this one)
  - Lecture 25: 04/22 (Tue)
  - Lecture 26: 04/24 (Thur)
  - Lecture 27: 04/29 (Tue)
  - Lecture 28: 05/06 (Tue)
- Quiz 5 (last quiz) during Lecture 26, 04/24
  - About oscillation

Recap: Oscillation

- Torsional pendulum
  - $\tau = -\kappa\phi$, where κ is torsion constant
  - $\dfrac{d^2\phi}{dt^2} + \dfrac{\kappa}{I}\phi = 0$
  - $\omega = \sqrt{\dfrac{\kappa}{I}}$
- Energy of oscillation: $E = \dfrac{1}{2}kA^2$

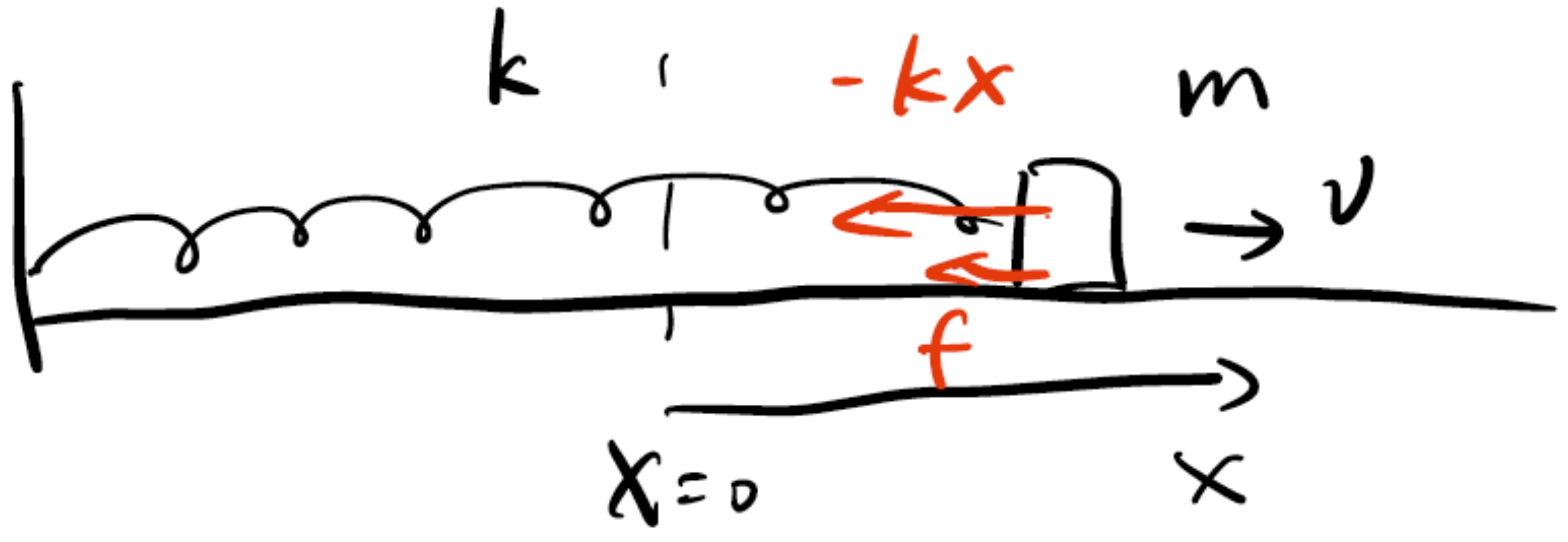
Recap: Damped Oscillation

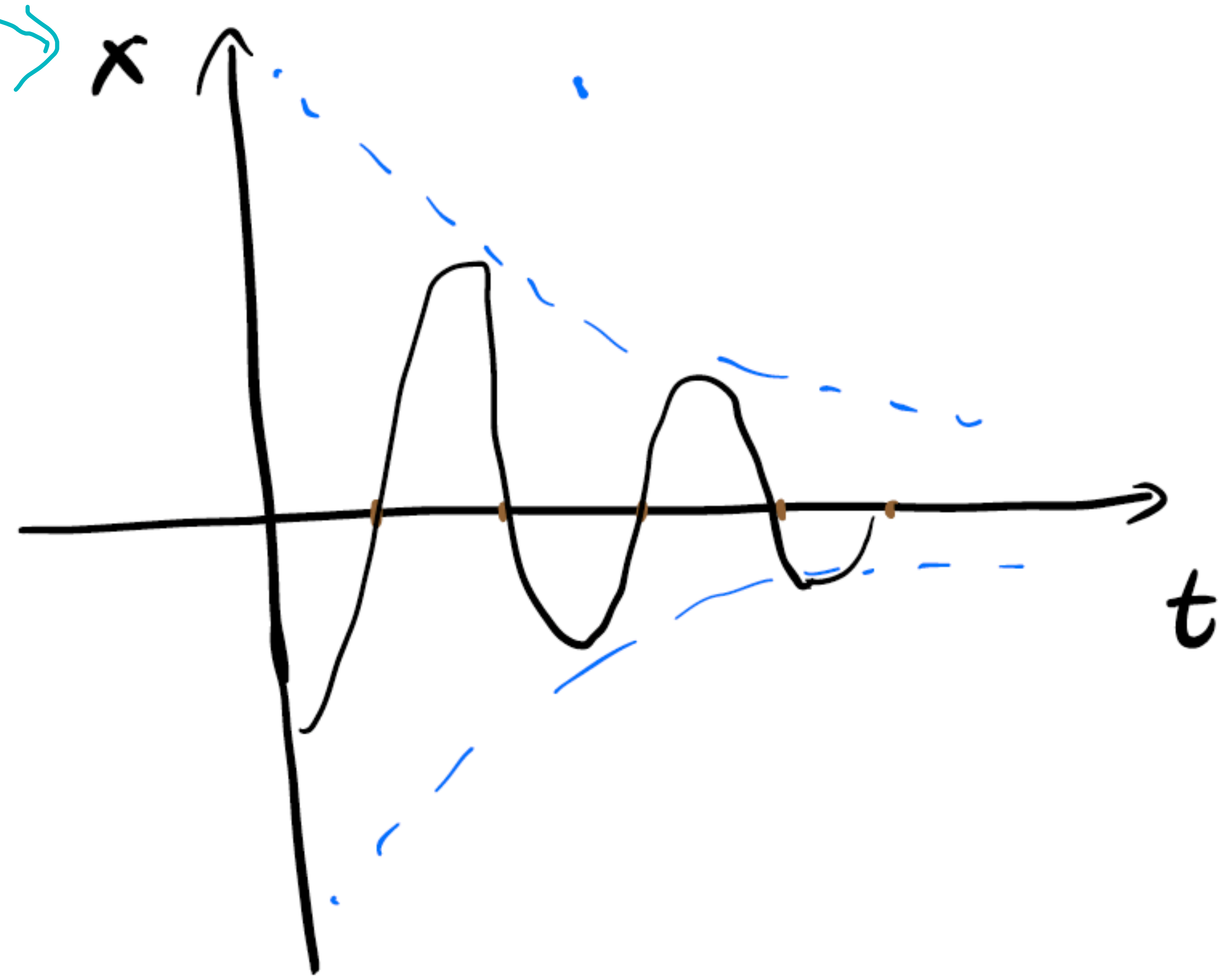
- Damped oscillation

  string force

  friction force

- $$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0$$
- $x(t) = A_0 e^{-\frac{\gamma}{2}t}\cos(\omega t + \phi)$, where
  - $\gamma = \frac{b}{m}$
  - $\omega = \omega_0\sqrt{1 - (\frac{\gamma/2}{\omega_0})^2}$, where $\omega_0 = \sqrt{k/m}$





- $Q = \frac{\omega_0}{\gamma}$

  large

  small

  - Assume $\frac{\gamma}{2} \ll w_0$ (underdamped)
  - $\frac{\Delta E}{E} = \frac{2\pi}{Q}$

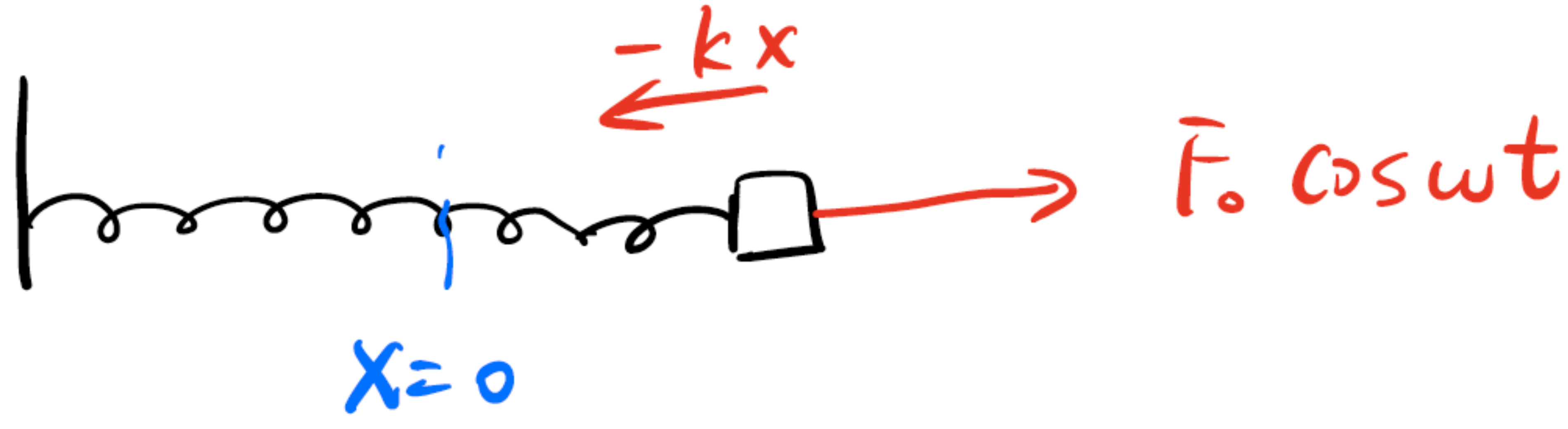
Recap: Forced Oscillation

- $ma = -kx + F_0 \cos \omega t$
- $\dfrac{d^2x}{dt^2} + \dfrac{k}{m}x = \dfrac{F_0}{m}\cos \omega t$ (general term)

Correction: free oscillaton term

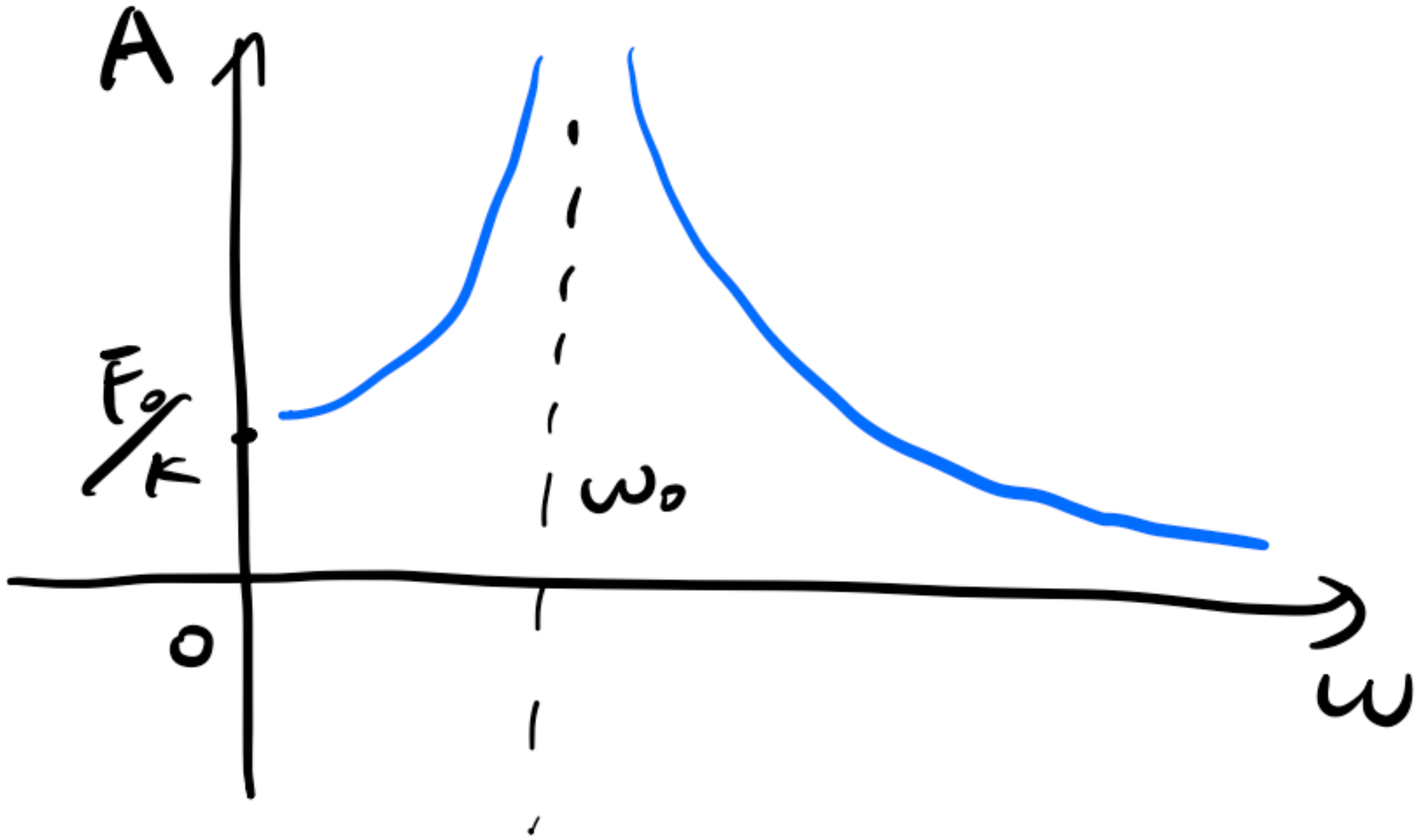
false oscillation (specific term)

- $x = \boxed{A_1 \cos(\omega_0 t + \phi_1)} + A_2 \cos(\omega t + \phi_2)$
  - First term: free oscillation
    - A_1 and ϕ_1 depend on the initial condition
  - Second term: forced oscillation
    - $A_2 = \dfrac{F_0/m}{|\omega_0^2 - \omega^2|}$
    - $\tan \phi = 0$ ($\phi = 0$ or π)

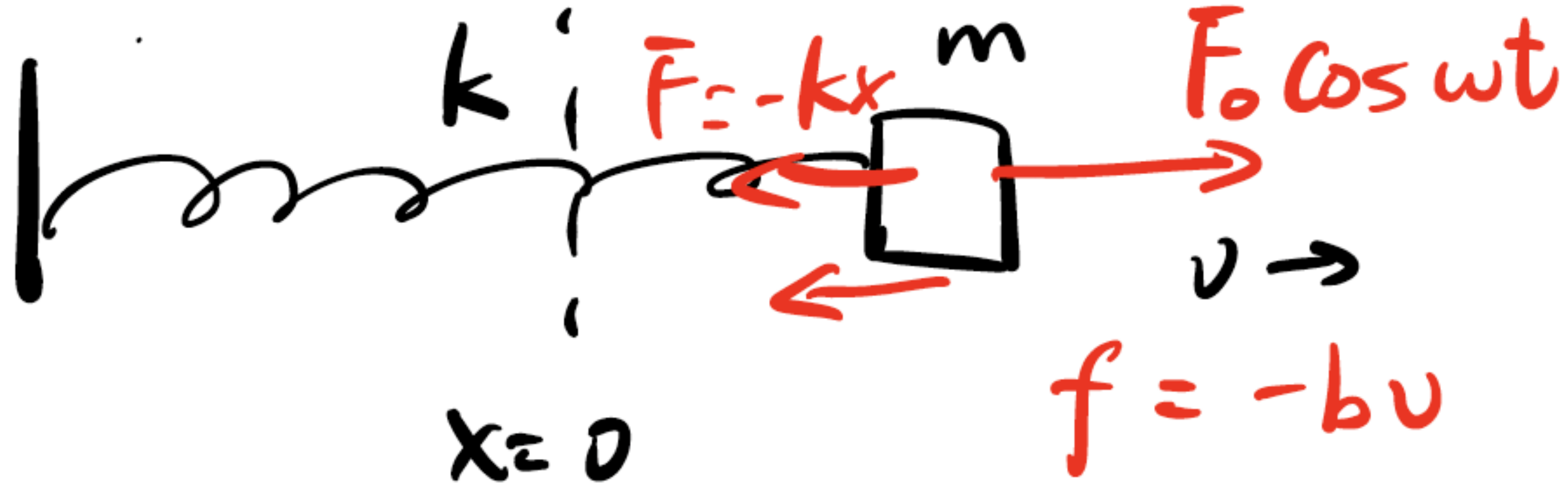


Recap: Forced Oscillation

- $A = \dfrac{F_0/m}{|\omega_0^2 - \omega^2|}$
  - $\omega \ll \omega_0$, $\phi = 0$, $A = F/k$
  - $\omega \gg \omega_0\,(\omega \to \infty)$, $\phi = \pi$, $A \to 0$
  - $\omega = \omega_0$, resonance, $A \to \infty$



Forced Oscillation w/ Damping



$\gamma = \frac{b}{m}$, $\omega_0 = \sqrt{\frac{k}{m}}$

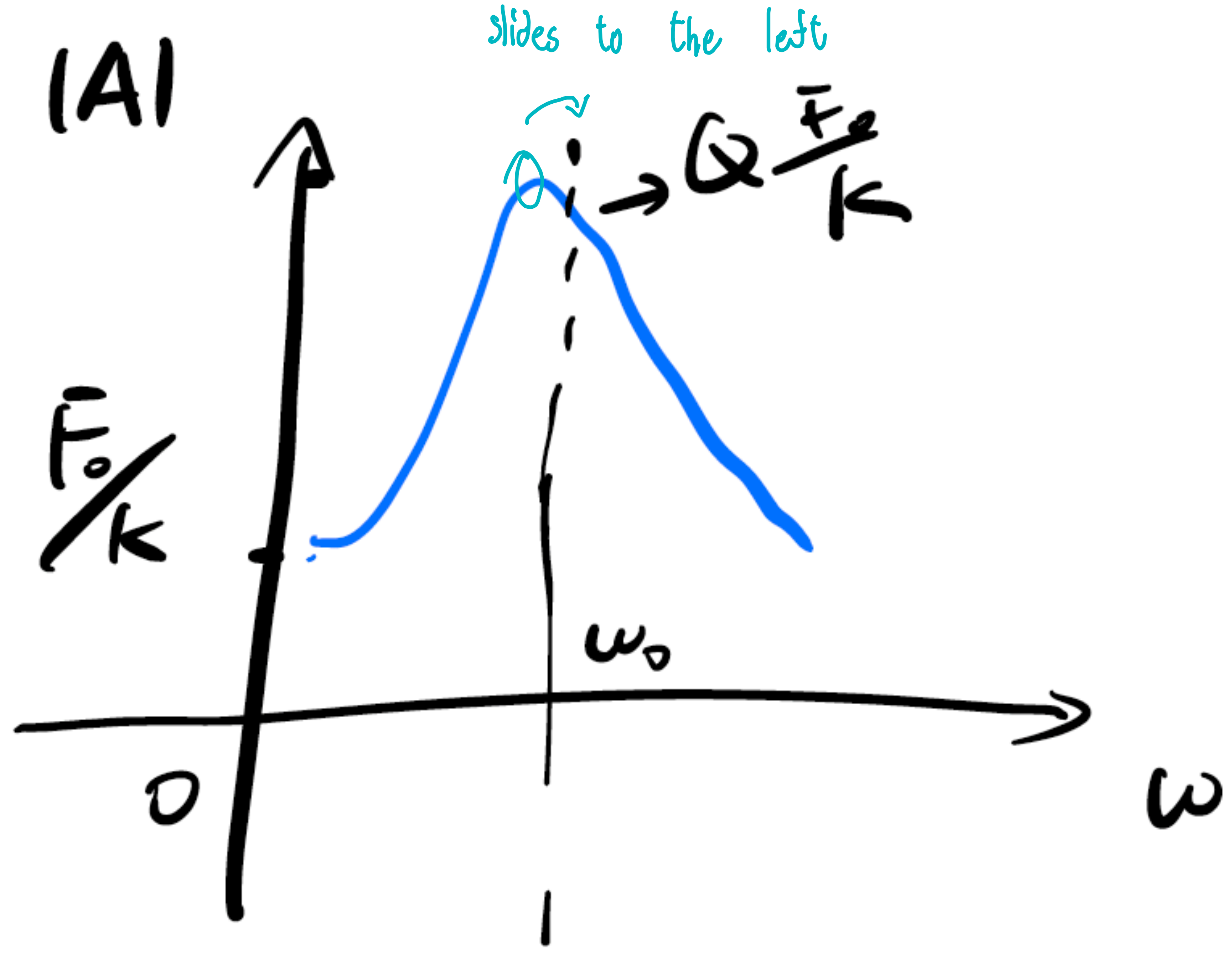
gamma

false (external force)

- $ma = -kx - bv + F_0 \cos \omega t$
- Equation of motion (EOM): $\frac{d^2x}{dt^2} + \omega_0^2 x + \gamma \frac{dx}{dt} - \frac{F_0}{m} \cos \omega t = 0$
- $x = A \cos(\omega t + \phi)$
  - $A = \frac{F/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\omega\gamma)^2}}$
  - $\tan \phi = -\frac{\omega\gamma}{\omega_0^2 - \omega^2}$

Forced Oscillation w/ Damping

- $$A = \frac{F/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\omega\gamma)^2}}$$
  - $\omega \ll \omega_0, A = F/k$
  - $\omega \gg \omega_0 \, (\omega \to \infty), A \to 0$
  - $\omega = \omega_0$, resonance, $A = \dfrac{F/m}{\omega_0 \gamma} = Q\dfrac{F}{k}$
- At which ω, A achieves the maximum?



large Q → ↓ damping

$$\frac{dA(w)}{dw} = 0 \quad , \quad \frac{d^2A}{dw^2} < 0$$

$$m\frac{d^2x}{dt^2} = -\upsilon' \frac{dx}{dt}$$

take as 0 → x

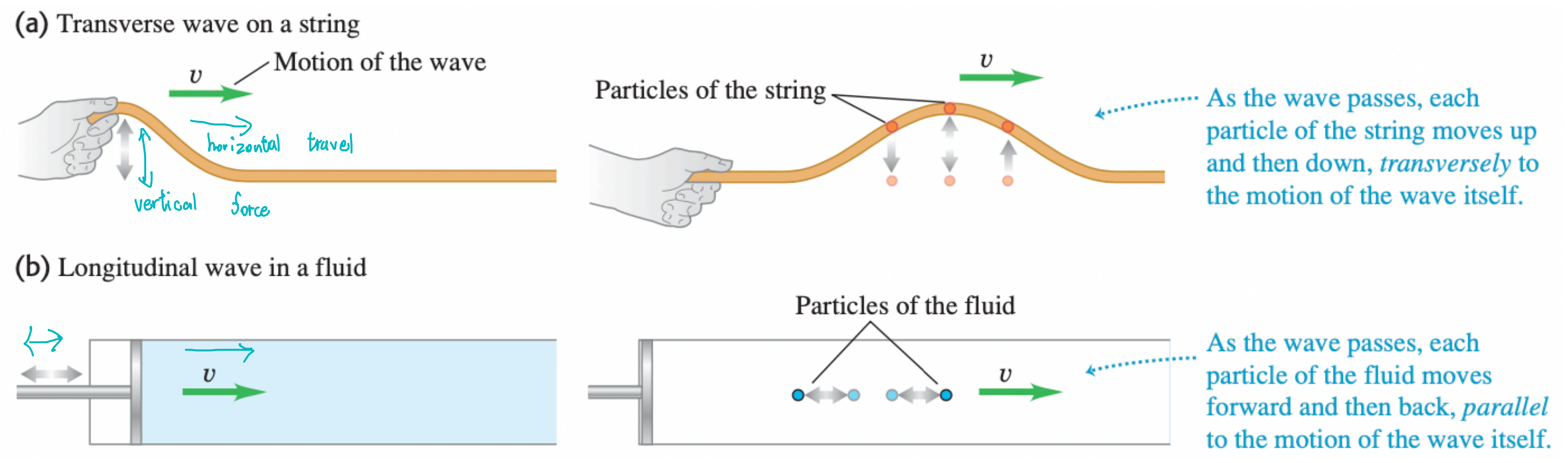
Wave

Outline

- General introduction
- Wave speed
- Wave function
- Wave equation
- Wave speed & equation for transverse wave
- Energy & power for transverse wave

Wave

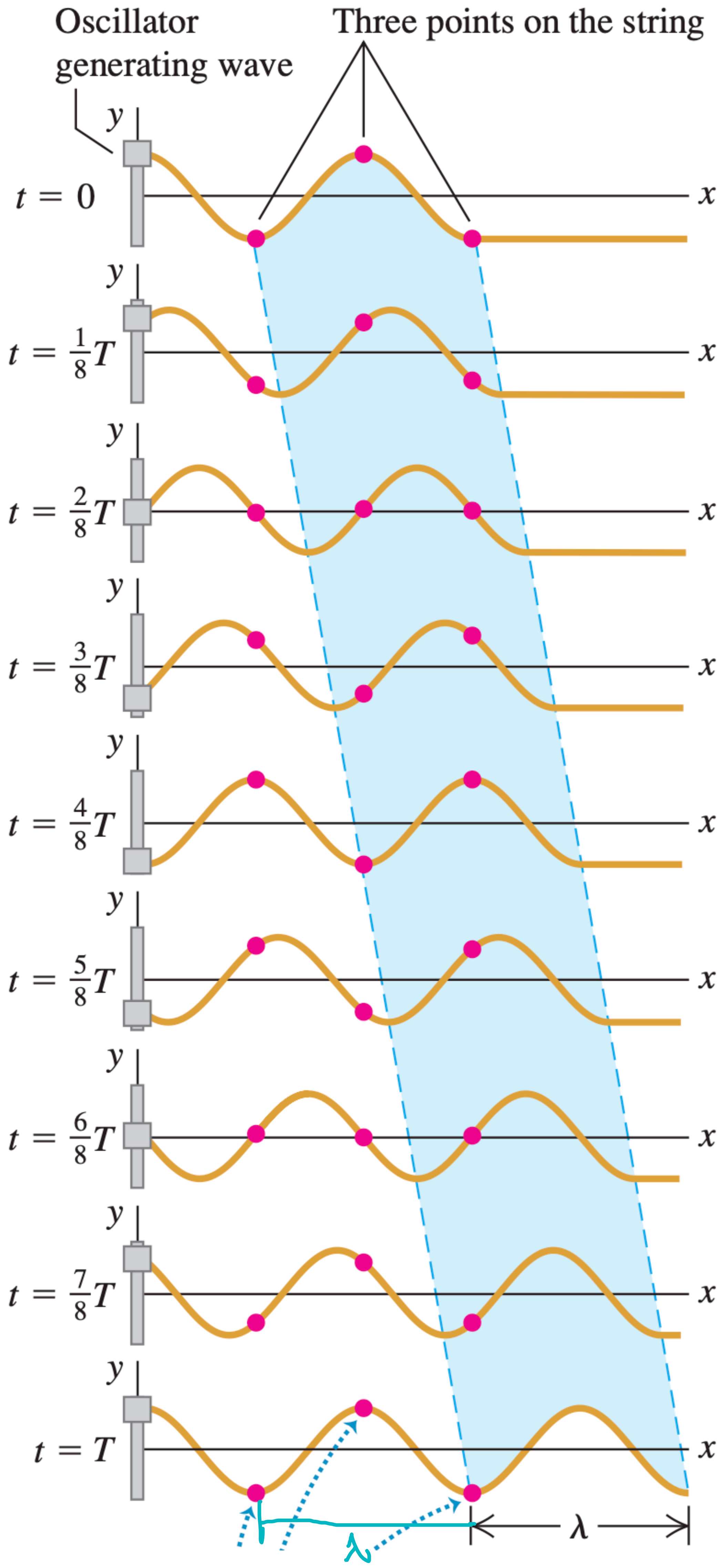
- Wave: An oscillation that travels in a medium.
- Two types:
  - Transverse wave: Oscillation direction is perpendicular to travel/propagation direction
  - Longitudinal wave: Oscillation direction is parallel to travel/propagation direction



Three Features of Wave

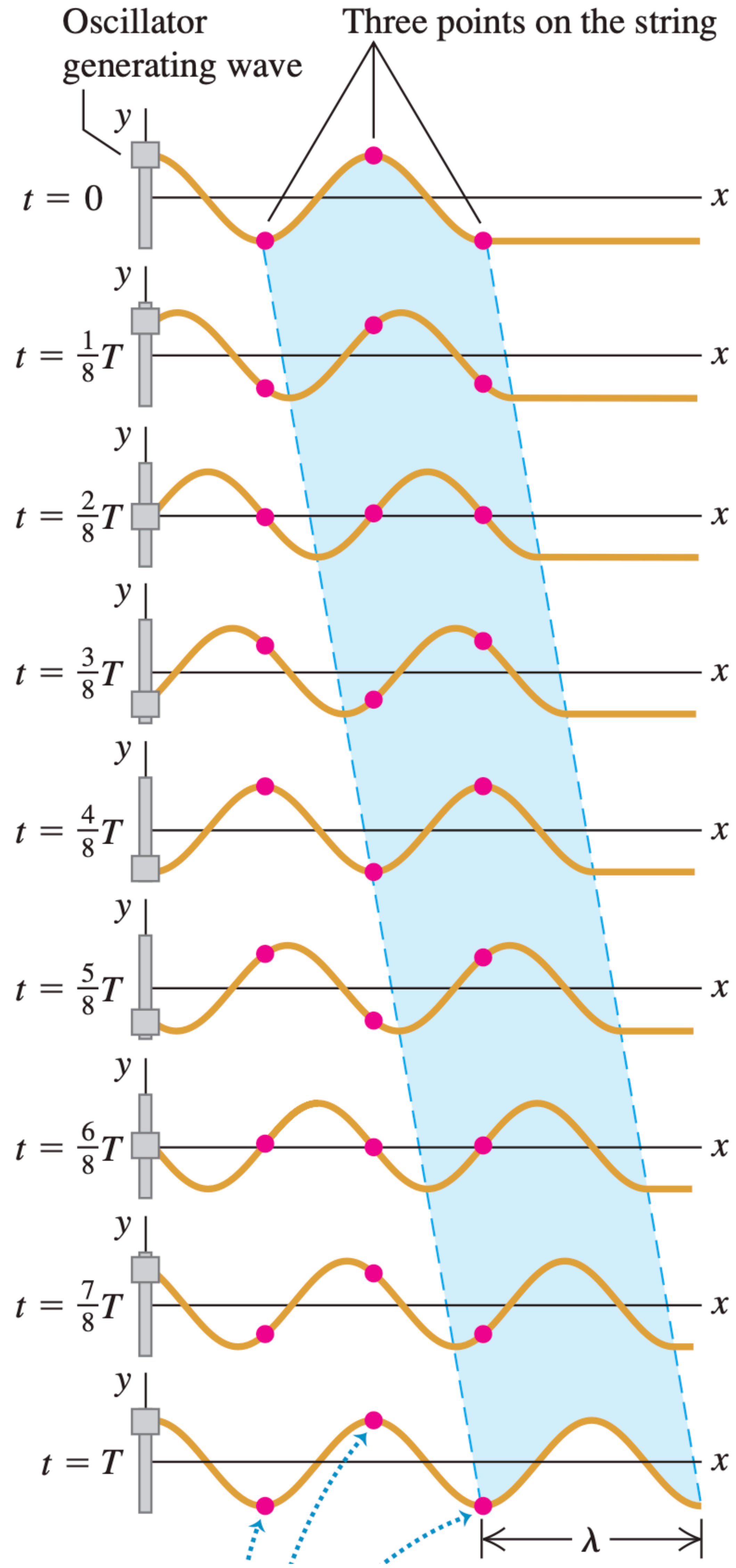
- Disturbance travels or propagates at a certain speed through the medium, which is called the wave speed or speed of propagation v.
- The medium does not travel, but oscillates around its equilibrium position. What travels is the overall pattern of disturbance.
- Energy is needed for wave propagation. Wave transports energy, but not matter.

Wave Speed



- Propagation/wave speed $v = \dfrac{\lambda}{T} = \lambda f$, where
  - λ: wavelength
  - T: period
  - $f = \dfrac{1}{T}$, frequency

Wave Function



- Wave function $y(x,t) = A\cos(kx - \omega t)$, where
  - A: amplitude (peak-to-equilibrium distance)
  - $k = \dfrac{2\pi}{\lambda}$, (angular) wave number
  - $kx - \omega t$: phase
- Wave speed $v = \dfrac{\lambda}{T} = \dfrac{\omega}{k}$

$\frac{2\pi}{k}$

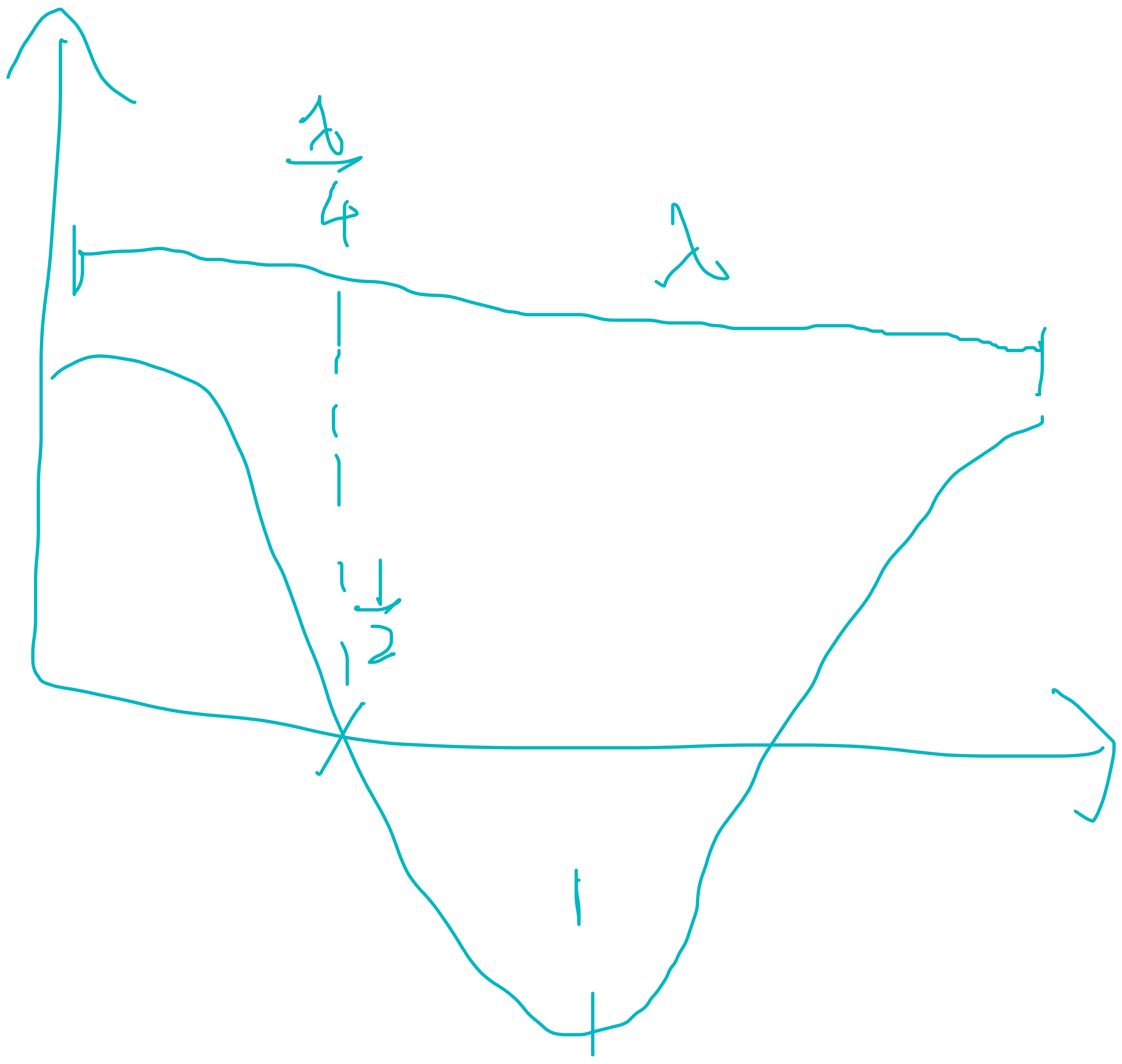
$\frac{2\pi}{\omega}$

- Travel direction
  - $y(x,t) = A\cos(kx - \omega t)$, moves in +x direction $= A\cos\left(k\left(x - \frac{\omega}{k}t\right)\right)$
  - $y(x,t) = A\cos(kx + \omega t)$, moves in -x direction $= A\cos(k(x - vt))$

$t = 0, \quad x = 0$

$t = t, \quad x = x + vt$

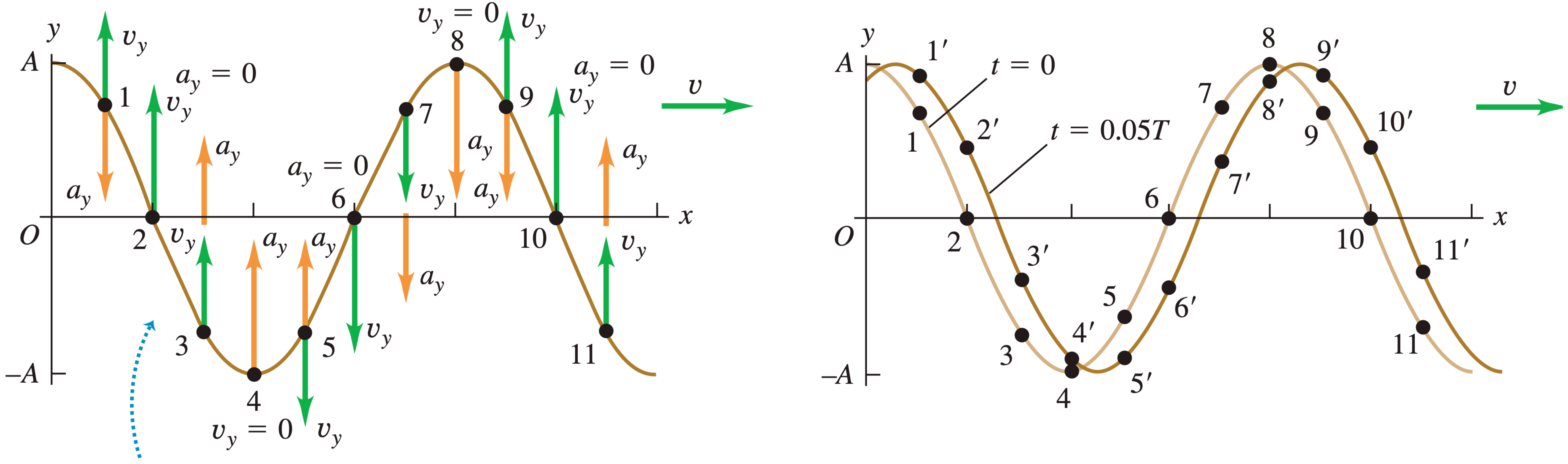
Particle Velocity & Acceleration for Transverse Wave



Particle Velocity & Acceleration

(a) Wave at $t = 0$

(b) The same wave at $t = 0$ and $t = 0.05T$

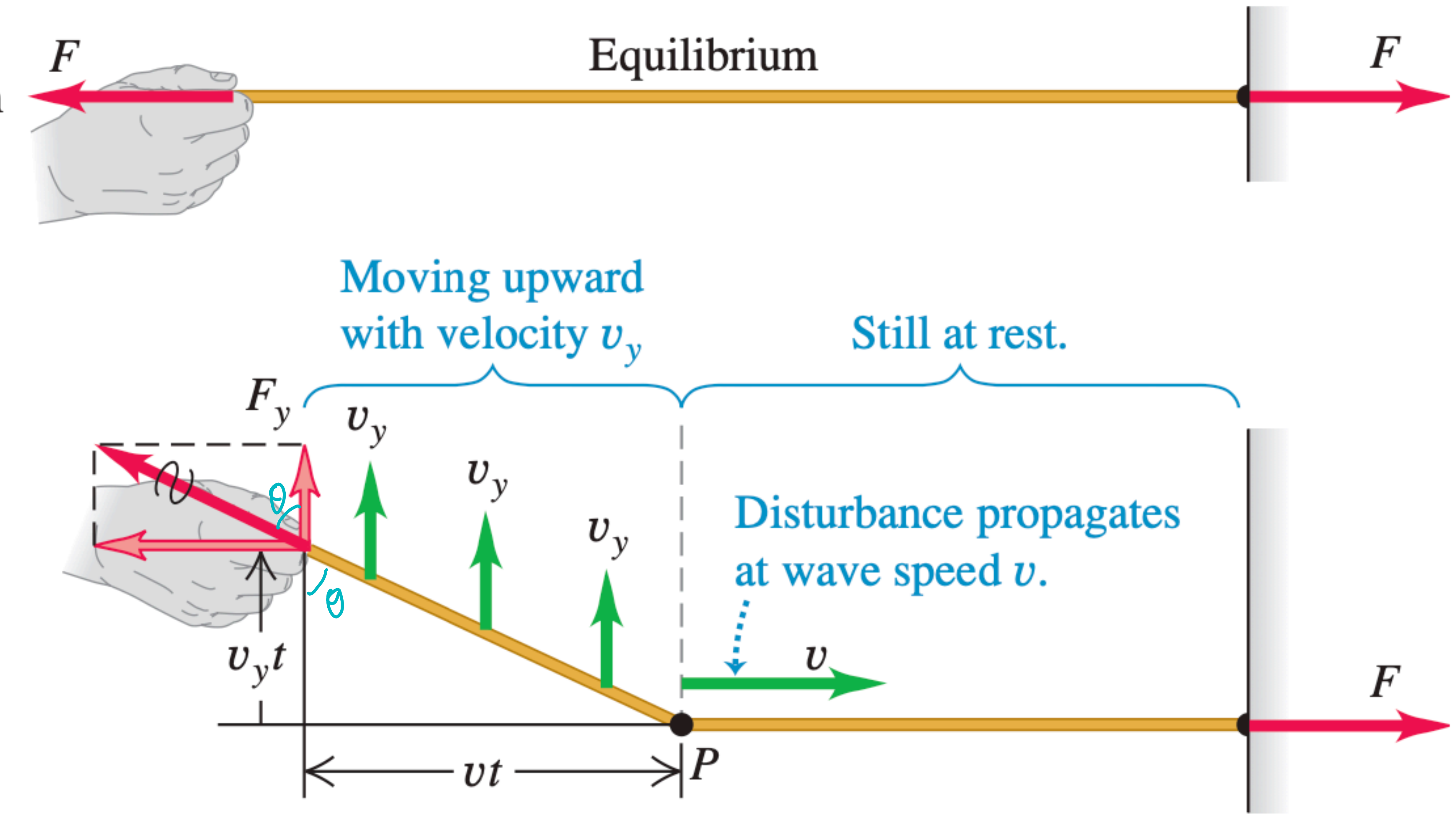


Wave Equation

- Wave equation: $\dfrac{\partial^2 y}{\partial t^2} = v^2 \dfrac{\partial^2 y}{\partial x^2}$

$y(x, t) = A\cos(kx - \omega t)$

Wave Speed of Transverse Wave



- We know:
  - μ: mass per unit length, kg/m
  - F_T: tension in the string
- Find wave speed v
- Idea: Impulse-momentum

vertical

$$J = \Delta p$$

$$J = F_y \cdot t$$

$$= F_T \cdot \frac{v_y}{v} \cdot t$$

$$\cos\theta \approx \frac{v_y t \leftarrow \text{ver}}{vt \leftarrow \text{hor}} = \frac{v_y}{v}$$

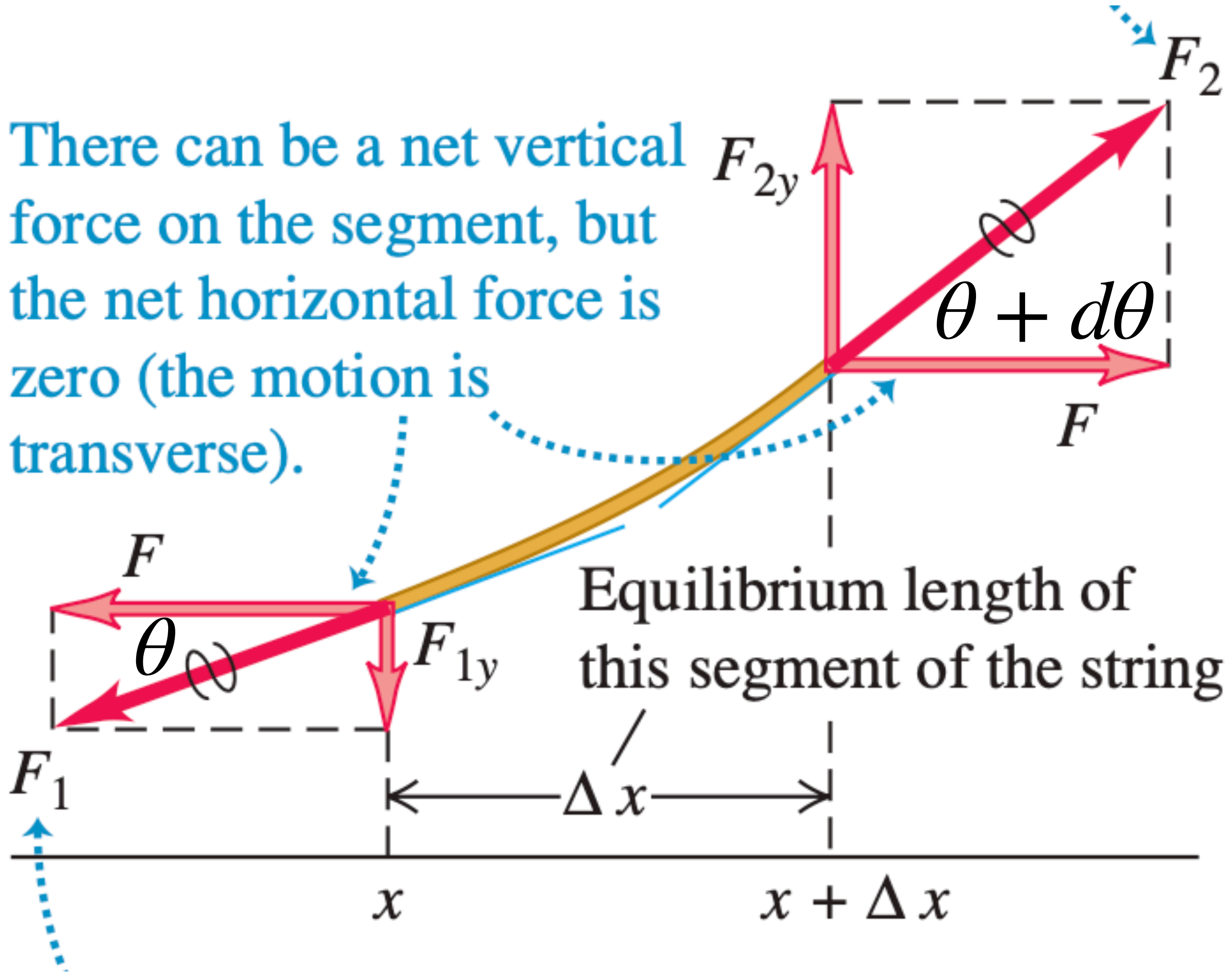
$$\Delta p = \mu \cdot vt \cdot v_y$$

$$F_T \cdot \frac{v_y}{v} t = \mu v t v_y$$

$$v = \sqrt{\frac{F_T}{\mu}} = \frac{\lambda}{T} = \frac{\omega}{k}$$

↑ transverse wave

Wave Function of Transverse Wave

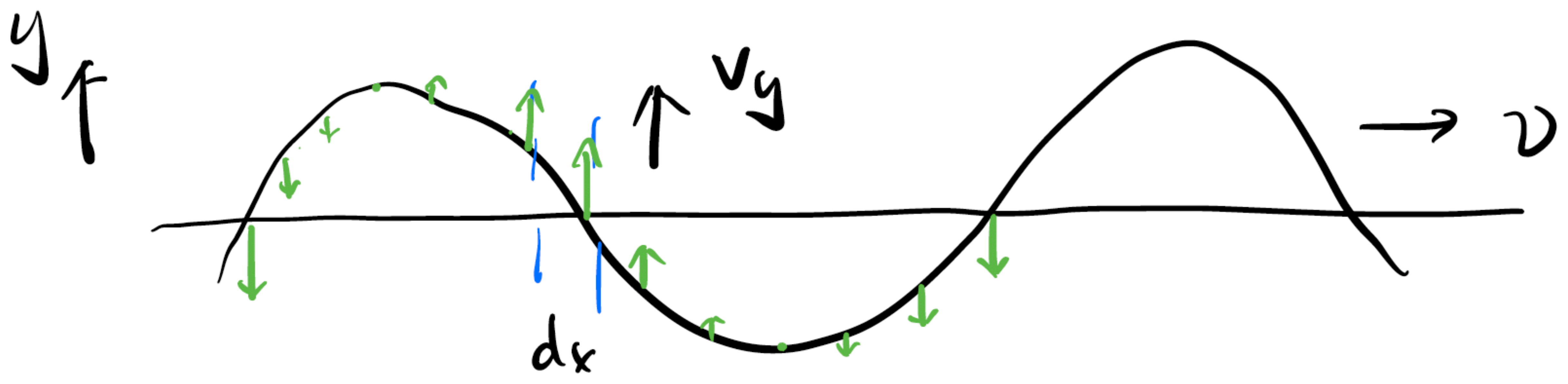


- We know F_T, μ
- Assumption: amplitude is small, meaning that $\theta \to 0$
- Newton's 2nd law

Wave Speed & Wave Function of Transverse Wave

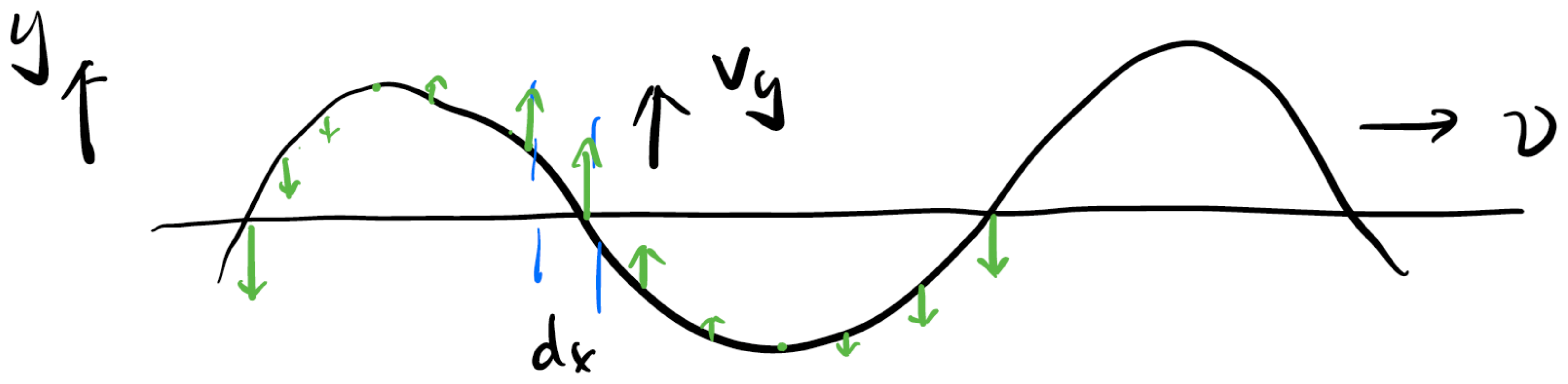
- Wave equaton for a mechanical transverse wave: $\dfrac{\partial^2 y}{\partial t^2} = \dfrac{F_T}{\mu}\dfrac{\partial^2 y}{\partial x^2}$, where
  - μ: mass per unit length, kg/m
  - F_{T}: tension in the string
- Wave speed $v = \sqrt{\dfrac{F_{\mathrm{T}}}{\mu}}$

Energy & Power in Traveling Wave



- We know
  - $y(x,t) = A\sin(kx - \omega t)$
  - $v = \sqrt{\dfrac{F_{\mathrm{T}}}{\mu}}$
- Find E in one wavelength.
- Since E=K+U, let's find K in one wavelength..

Energy & Power in Traveling Wave



- We know
  - $y(x,t) = A\sin(kx - \omega t)$
  - $v = \sqrt{\dfrac{F_{\mathrm{T}}}{\mu}}$
- Find E in one wavelength.
- Since E=K+U, let's find U in one wavelength..

Energy & Power in Traveling Wave

- Total energy in one wavelength $E = \pi\sqrt{\mu F_{\mathrm{T}}}\omega A^2$
  - $K = \dfrac{\pi}{2}\sqrt{\mu F_{\mathrm{T}}}\omega A^2$
  - $U = \dfrac{\pi}{2}\sqrt{\mu F_{\mathrm{T}}}\omega A^2$
- Average power $P_{\mathrm{av}} = \dfrac{E}{T} = \dfrac{1}{2}\sqrt{\mu F_{\mathrm{T}}}\omega^2 A^2$